Corn Foliar Disease Management

Covers: Gray Leaf Spot, Common Rust, Northern Corn Leaf Blight

Compiled from public university cooperative extension publications for testing InsightAI-RAG's retrieval and grounding quality against real agricultural guidance. This is not a substitute for consulting a local agronomist or extension office before applying any treatment. Always follow current product labels; pesticide registrations change and vary by region.

Gray Leaf Spot (GLS) — Cercospora zeae-maydis

One of the most common foliar diseases of corn in humid regions, especially with reduced tillage. Lesions first appear on lower leaves as yellow or tan rectangular spots with darkened margins, bounded by leaf veins; as they mature they elongate, turn gray, and sporulate. GLS rarely affects yield severely but can be significant on susceptible hybrids under high disease pressure, high humidity, low-lying fields, and no-till/conservation tillage.

Management: Destroy crop debris after harvest and till it as deep as possible (burning alone is not effective). Rotate away from corn — a 3-year rotation is more effective than 2-year, which is better than none, since the fungus survives on corn residue. Plant hybrids with partial GLS resistance. Reduce drought and nutrient stress. Scout lower leaves before tasseling (VT); fungicide applications between VT and R2 can protect yield when disease pressure is high.

Source: NC State Extension Publications, 'Gray Leaf Spot in Corn' (Corn Disease Information series, reviewed 2023)

Common Rust — Puccinia sorghi

Begins as fleck-like lesions developing into small, tan, circular-to-elongate spots on both leaf surfaces, erupting into jagged brick-red to cinnamon-brown pustules that readily rub off on fingers; darker orange-brown teliospores appear as the disease progresses. Favored by cool, moist conditions (61-77F with 6+ hours of dew). Generally far less destructive than southern rust.

Management: The most cost-effective control is planting rust-resistant hybrids where available. Spray when 80% of observed leaves have one or more pustules and more than 1-2% of leaf area is affected; spraying corn within two weeks of black layer, or with yield potential under 100 bu/acre, is unlikely to pay for itself.

Source: NC State Extension Publications, 'Corn Rusts: Common and Southern Rust' (reviewed 2023)

Northern Corn Leaf Blight (NCLB) — Exserohilum turcicum

Can cause 30-50% yield loss in dent corn if established before tasseling, plus quality loss in sweet and silage corn. Early symptoms are long, narrow tan lesions parallel to leaf veins; these expand into diagnostic long, oblong 'cigar-shaped' tan-to-gray lesions that coalesce over time. Favored by moderate temperatures (64-84F) and 6-18 hours of leaf wetness — cooler and wetter conditions than gray leaf spot prefers.

Management: Plant hybrids with NCLB resistance ratings, especially in fields with disease history. Rotate away from corn for 1-2 years (soybeans, small grains, vegetables are good rotation crops; avoid sorghum/sudangrass, which can host the same pathogen). Tillage buries and speeds decomposition of infected residue — clean equipment before moving between fields. Fungicide applications between VT and R3 have shown the greatest efficacy when conditions remain favorable for disease.

Source: University of Delaware Cooperative Extension Fact Sheet, 'Northern Corn Leaf Blight' (Oct 2025)

Healthy

No signs of disease detected. Continue scouting lower leaves through the season, especially in fields with a history of foliar disease or reduced tillage.

Source: General extension guidance